Test pyPDFCodebook

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Project Overview: Summary of Project Details

Project Title: PDF Codebook with Sample Data

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Project Description : pyPDFCodeBook helps researchers and data professionals create clear, attractive codebooks for tabular datasets. Codebooks document essential metadataâ€"project descriptions, data provenance, variable definitions, and summariesâ€"ensuring your data is understandable and reproducible. While datasets contain values, they rarely explain what each column or row represents. Codebooks fill this gap by providing structured, self-explanatory documentation. They reinforce best practices such as tidy data, unique keys, and transparent variable origins. With pyPDFCodeBook, generating a professional, easy-to-read codebook takes just a few clicksâ€"laying the foundation for good data science and reproducible research. Remember: You are your number one data user, so help your future self out and document your metadata.

Data Provenance : All Regional Satisfaction Survey sample data generated using Claude Sonnet 4 in VS Code Agent mode. All values are random and do not represent actual survey data. Logo image generated using MS Copilot.

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Center for Risk-Based Community Resilience Planning : NIST-70NANB20H008, NIST-70NANB15H044 (http://resilience.colostate.edu/)

Southeast Texas Urban Integrated Field Lab: DE-SC0023216 (https://setx-uifl.org/)

Related Works:

Keywords: Codebook, data dictionary, metadata

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Data Dictionary: Summary of Variables

Variable Name	Data Type	Length	Categorical	Variable Label
rid	String			Random Survey Response ID
region	Int		True	Geographic Region
satscore	Int		True	Satisfaction Rating
weight	Float			Survey Weight
age	Int			Age in Years

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Variable Details and Notes

The following pages provide details on each variable. Where applicable, notes provide links to verify data. Categorical variables include details on category codes.

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rid: Random Survey Response ID

Variable characteristic	Variable details
variable type	string
total cases	1,000
valid cases	1,000
missing cases	0
unit of measure	Responses
unit of analysis	Survey response
unique values	1,000
minimum length	5
maximum length	5
example 1	M1101
example 2	E1526
example 3	P1826
example 4	S1226

Variable Notes: rid

1. Unique, non-missing key for sample data.
2. Randomly generated 5 digit alphanumeric string identifier.
3. Range from A1000 to Z9999.

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region: Geographic Region

Variable characteristic	Variable details
variable type	categorical (Int)
categorical type	nominal
total cases	1,000
valid cases	1,000.0
missing cases	0
weighted total cases	4,700.00
unit of measure	Responses by Region
unit of analysis	Survey response
range	minimum value: 1 to maximum value: 4

region: Geographic Region - Categorical codes, labels and frequencies

Code	Label	Count of Responses	Percent Responses	Weighted Sum	Weighted Percent
1	1. North	200	20.00%	2,000.0	42.55%
2	2. South	300	30.00%	1,200.0	25.53%
3	3. East	400	40.00%	300.0	6.38%
4	4. West	100	10.00%	1,200.0	25.53%

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satscore: Satisfaction Rating

Variable characteristic	Variable details
variable type	categorical (Int)
categorical type	ordinal
total cases	1,000
valid cases	900.0
missing cases	100
weighted total cases	4,222.25
unit of measure	Satisfaction Level
unit of analysis	Survey response
range	minimum value: 1 to maximum value: 5
mean	3.67
weighted mean	3.75

satscore: Satisfaction Rating - Categorical codes, labels and frequencies

Code	Label	Count of Responses	Percent Responses	Weighted Sum	Weighted Percent
1	1. Very Dissatisfied	38	4.22%	173.75	4.12%
2	2. Dissatisfied	110	12.22%	443.0	10.49%
3	3. Neutral	174	19.33%	755.25	17.89%
4	4. Satisfied	364	40.44%	1,725.75	40.87%
5	5. Very Satisfied	214	23.78%	1,124.5	26.63%

Variable Notes: satscore

1. Five-point Likert scale measuring overall satisfaction.
2. Missing values indicate participant did not respond to this question.

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weight: Survey Weight

Variable characteristic	Variable details
variable type	numeric (Float)
total cases	1,000
valid cases	1,000
missing cases	0
unit of measure	Population
unit of analysis	Survey response
range	minimum value: 0.75 to maximum value: 12.00
mean	4.70
median	4.00
standard deviation	4.17
10th percentile	0.75
25th percentile	0.75
50th percentile	4.00
75th percentile	10.00
90th percentile	10.20

Variable Notes: weight

1. Statistical weight for population inference and representative results.
2. Used to adjust for sampling bias and ensure results reflect target population.
3. Based on sample design for each region.
4. Region 1 weight = 10.0, Region 2 weight = 4.0, Region 3 weight = 0.75, Region 4 weight = 12.0.
5. Regions 1, 2, and 4 are under sampled regions. Region 3 is over sampled.

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age: Age in Years

Variable characteristic	Variable details
variable type	numeric (Int)
total cases	1,000
valid cases	1,000
missing cases	0
unit of measure	Years
unit of analysis	Survey response
range	minimum value: 18.00 to maximum value: 99.00
mean	47.03
median	37.00
standard deviation	21.86
10th percentile	23.00
25th percentile	32.00
50th percentile	37.00
75th percentile	59.00
90th percentile	81.00

Variable Notes: age

1. Age of survey respondent at time of survey.

2. Range from 18 to 99 years.

3. Missing values indicate participant did not provide age information.4. Surveys completed between November 2025 and December 2025.

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Key Terms and Definitions

Codebook: "A codebook describes the contents, structure, and layout of a data collection... The main body of a codebook contains unambiguous variable level details. " (ICPSR: https://www.icpsr.umich.edu/sites/icpsr/posts/shared/what-is-a-codebook).

Data Dictionaries: A compact list of the variables in the dataset. Includes variable name, data type, length, if the variable is categorical or not, and the variable label.

Key: "A key is a variable or set of variables that uniquely identifies the elements of a table. The variables that form the key never take on missing values, and a keyâ€™s value is never duplicated across rows of the table." (Gentzko & Shapiro (2014). Code and Data for the Social Sciences: A Practitionerâ€™s Guide. p. 20 https://web.stanford.edu/~gentzkow/research/CodeAndData.pdf)

Metadata: "Metadata is information about the data, descriptors that facilitate cataloguing data and data discovery." (The Turing Way. "Metadata in Research Data Management." The Turing Way Book: https://book.the-turing-way.org/reproducible-research/rdm/rdm-metadata).

Provenance: "The precise subset and version of data a set of result originates from. (The Turing Way Book: https://book.the-turing-way.org/reproducible-research/vcs/vcs-data/).

Tabular Dataset: "Datasets made up of rows and columns." (Tidy data: https://tidyr.tidyverse.org/articles/tidy-data.html)

Tidy Data: "Each variable is a column; each column is a variable. Each observation is a row; each row is an observation. Each value is a cell; each cell is a single value." (Tidy data: https://tidyr.tidyverse.org/articles/tidy-data.html)

Unit of Observation: "The object(s) of analysis for the data collection, such as an organization, individual, or household." (ICPSR: https://icpsr.github.io/metadata/icpsr_study_schema/#unit_of_observation).